Exam in Signals & Transforms 2026-01-12

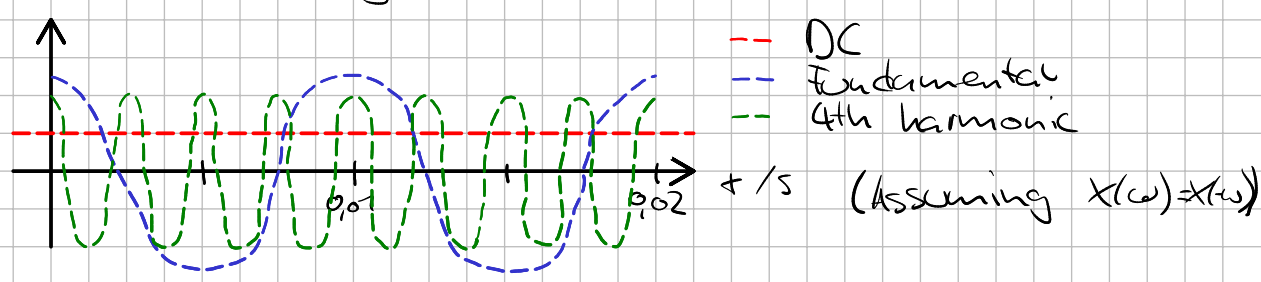
1.a) Two properties of the signal are:

- Periodic: The spectrum $X(\omega)$ is a line spectrum, which means that the signal is periodic, $x(t) = x(t+T_0)$
- Even: The spectrum is purely real (see the y-axis label), which implies that the signal must be even, $x(t) = x(-t)$.

b) From the spectrum, we can see that the signal has a non-zero DC-offset and is dominated by two components, the fundamental component with $\omega_0 = 200\pi$ rad/s and the 4th harmonic with $\omega = 4\omega_0 = 800\pi$ rad/s. The 4th harmonic has slightly lower amplitude than the fundamental. Furthermore, the fundamental period is given by

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{200\pi} = \frac{1}{100}\,\text{s} = 0{,}01\,\text{s}.$$

This yields something like the sum of these components:



(Assuming $X(\omega) = X(-\omega)$)

c)

$$y(t) = x(t) * 300\,\text{sinc}(300t)$$

$$\Updownarrow$$

$$Y(\omega) = X(\omega) \cdot 300\,\mathcal{F}\{\text{sinc}(300t)\}$$

$$\mathcal{F}\{\text{sinc}(300t)\} \underset{\uparrow}{=} \frac{1}{300}\,\text{rect}\left(\frac{\omega}{2\pi \cdot 300}\right) = \frac{1}{300}\,\text{rect}\left(\frac{\omega}{600\pi}\right)$$

$$\text{sinc}(t/T_0) \Leftrightarrow T_0\,\text{rect}(\omega T_0/2\pi)$$

$$Y(\omega) = X(\omega) \cdot 300 \cdot \frac{1}{300}\,\text{rect}\left(\frac{\omega}{600\pi}\right)$$

$$= X(\omega) \cdot \mathrm{rect}\left(\frac{\omega}{600\pi}\right)$$

For $\mathrm{rect}\left(\frac{\omega}{600\pi}\right)$ it holds that

$$\mathrm{rect}\left(\frac{\omega}{600\pi}\right) = \begin{cases} 1 & |\omega| \leq 300\pi, \\ 0 & \text{otherwise.} \end{cases}$$

Hence, $Y(\omega)$ only contains the DC and fundamental components of $X(\omega)$.

$$\Rightarrow y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} Y(\omega)\, e^{j\omega t}\, d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\frac{3\pi}{4}\delta(\omega + 200\pi) + \frac{\pi}{4}\delta(\omega) + \frac{3\pi}{4}\delta(\omega - 200\pi)\right] e^{j\omega t}\, d\omega$$

Note: This assumes $X(\omega) = X(-\omega)$ (i.e. real $x(t)$). Assuming $X(\omega) = 0$ for $\omega < 0$ (as stated) would give a complex $x(t)$. This solution is also accepted.

$$= \frac{1}{2\cancel{\pi}} \left[\frac{3\cancel{\pi}}{4} e^{-j200\pi t} + \frac{\cancel{\pi}}{4} + \frac{3\cancel{\pi}}{4} e^{j200\pi t}\right]$$

$$= \frac{3}{4}\cos(200\pi t) + \frac{1}{8}$$

$$\Rightarrow y(t) = \frac{1}{8} + \frac{3}{4}\cos(200\pi t) \quad \text{or} \quad y(t) = \frac{1}{8} + \frac{3}{8} e^{j200\pi t}$$

d) By using the Laplace transform's derivative property

$$\frac{d\,x(t)}{dt} \Leftrightarrow sX(s) - x(0)$$

and multiple applications thereof, we can transform the differential equation into a algebraic equation, for example (with zero initial conditions):

$$s^N a_N Z(s) + \ldots + s a_1 Z(s) + a_0 Z(s) = b_0 V(s) + s b_1 V(s) + \ldots + s^M b_M V(s)$$

This algebraic equation can then easily be solved and the result can be transformed back to the time domain. Hence, solving the differential equation becomes easier.

2.a) The impulse response is given by the inverse laplace transform of the transfer function: $h(t) \Leftrightarrow H(s)$.

We can rewrite $H(s)$ as follows:

$$H(s) = \frac{4}{s^2 + 0{,}2s + 2} \underset{\text{Completing the square}}{=} \frac{4}{s^2 + 0{,}2s + (0{,}1)^2 - (0{,}1)^2 + 2}$$

$$= \frac{4}{(s+0{,}1)^2 + 1{,}99} = K \frac{\omega_0}{(s+a)^2 + \omega_0^2}$$

where

$$\omega_0^2 = 1{,}99 \Rightarrow \omega_0 = \sqrt{1{,}99} = 1{,}41$$

$$a = 0{,}1$$

$$K\omega_0 = 4 \Rightarrow K = \frac{4}{\omega_0} = 2{,}84$$

$$H(s) = 2{,}84 \frac{1{,}41}{(s+0{,}1)^2 + (1{,}41)^2}$$

$$\Updownarrow$$

$$\underline{\underline{h(t) = 2{,}84\, e^{-0{,}1t} \sin(1{,}41\, t)\, u(t)}}$$

b) If a bounded $x(t)$ shall give a bounded output $y(t)$, the system needs to be BIBO-stable. This can be checked by checking the system's poles:

$$s^2 + 0{,}2s + 2 = 0$$

pq-formula:

$$p_{1,2} = -\frac{0{,}2}{2} \pm \sqrt{\left(\frac{0{,}2}{2}\right)^2 - 2}$$

$$= -0{,}1 \pm \sqrt{0{,}01 - 2}$$

$$= -0{,}1 \pm \sqrt{-1{,}99}$$

$$p_{1,2} = -0{,}1 \pm j\,1{,}41$$

$$\approx -0{,}1 \pm j\,1{,}41$$

For the system to be stable, it must hold that $\mathrm{Re}\{p_i\} < 0 \;\forall\, p_i$, which is the case here. Hence, we have that the system is stable and will yield a bounded output $y(t)$.

c) As shown in b), the system is a second order system with a complex conjugate pole pair. Rewriting to Bode form:

$$H(s) = \frac{4}{s^2 + 0{,}2s + 2} = \frac{4}{2\,(s/2 + 0{,}1s + 1)} = \frac{2}{s/2 + 0{,}1s + 1}$$

For the second order term in the denominator, we can identify:

$$\omega_n^2 = 2 \Rightarrow \omega_n = \sqrt{2} \approx 1{,}41$$

$$2\frac{\xi}{\omega_n} = 0{,}1 \Rightarrow \xi = \frac{0{,}1\,\omega_n}{2} = \frac{0{,}1\sqrt{2}}{2} = \frac{0{,}1}{\sqrt{2}} \approx 0{,}071$$

$$\Rightarrow H(s) = \frac{2}{s/2 + 2\frac{0{,}071}{\sqrt{2}}s + 1}$$

Hence, we have 2 terms:

1. A constant in the numerator, $K = 2$

   $\Rightarrow K_{dB} = 20\log_{10}(2) = 6\ \mathrm{dB}$

2. A second order term $\frac{1}{s/2 + 2\frac{0{,}071}{\sqrt{2}}s + 1}$ in the denominator:

   - $\omega \ll \omega_n$: 0 dB

- $\omega \gg \omega_n$: -40 dB/decade
- $\omega = \omega_n = \sqrt{2}$: Peak with amplitude

$$-20 \log_{10}\left(\sqrt{\left(1-\frac{\omega}{\sqrt{2}}\right)^2 + \left(2\frac{0{,}071}{\sqrt{2}}\omega\right)^2}\right)$$

$$= -20 \log_{10}(2 \cdot 0{,}071) \approx 17 \text{ dB}$$

$\Rightarrow$ See the Bode diagram at the end.

d) We can use the laplace-domain input-output relationship:

$$Y(s) = H(s)\,U(s)$$

with

$$U(s) = \mathcal{L}\{u(t)\} = \frac{1}{s}$$

$$\Rightarrow Y(s) = \frac{4}{s^2+0{,}2s+2} \cdot \frac{1}{s} = \frac{As+B}{s^2+0{,}2s+2} + \frac{C}{s}$$

Partial fraction expansion

$$= \frac{s(As+B) + C(s^2+0{,}2s+2)}{s(s^2+0{,}2s+2)}$$

Numerator only:

$$s(As+B) + C(s^2+0{,}2s+2) = 4$$

$$As^2 + Bs + Cs^2 + 0{,}2Cs + 2C = 4$$

$$s^2(A+C) + s(B+0{,}2C) + 2C = 4$$

Coefficient comparison:

s^0: $2C = 4$

$\underline{C = 2}$ (1)

s^1: $B + 0{,}2C = 0$

$B + 0{,}4 = 0$

$$\underline{B = -0{,}4} \qquad (2)$$

$$s^2: \quad A + C = 0$$

$$A + 2 = 0$$

$$\underline{A = -2} \qquad (3)$$

Hence, we have:

$$Y(s) = \frac{-2s - 0{,}4}{s^2 + 0{,}2s + 2} + \frac{2}{s} = \underbrace{\frac{2}{s}}_{1} - \underbrace{\frac{2s + 0{,}4}{s^2 + 0{,}2s + 2}}_{2}$$

First term 1: $\frac{2}{s} \Leftrightarrow 2u(t)$

Second term 2: Rewrite to the form

$$K_1 \frac{s+a}{(s+a)^2 + \omega_0^2} + K_2 \frac{\omega_0}{(s+a)^2 + \omega_0^2}$$

From a), we have $a = 0{,}1$ and $\omega_0 = \sqrt{1{,}99}$. Hence, we have

$$K_1(s+a) + K_2\omega_0 = 2s + 0{,}4$$

$$K_1 s + K_1 a + K_2 \omega_0 = 2s + 0{,}4$$

$$\Rightarrow \quad K_1 s = 2s \quad \Rightarrow \quad K_1 = 2$$

$$\Rightarrow \quad K_1 a + K_2 \omega_0 = 0{,}4$$

$$K_2 = \frac{0{,}4 - K_1 a}{\omega_0} = \frac{0{,}4 - 2 \cdot 0{,}1}{\sqrt{1{,}99}}$$

$$\approx 0{,}14$$

Thus, we have for 2:

$$2 \frac{s + 0{,}1}{(s+0{,}1)^2 + 1{,}99} + 0{,}14 \frac{\sqrt{1{,}99}}{(s+0{,}1)^2 + 1{,}99}$$

$$\Updownarrow$$

$$2 e^{-0{,}1t} \cos(1{,}41t) u(t) + 0{,}14 e^{-0{,}1t} \sin(1{,}41t)$$

$\Rightarrow$ Finally:

$$\underline{\underline{y(t) = \left(2 - 2e^{-0{,}1t}\cos(1{,}41t) - 0{,}14 e^{-0{,}1t} \sin(1{,}41t)\right) u(t)}}$$

3.a) The filter order for a Chebyshev II lowpass filter is given by:

$$N \geq \frac{\operatorname{arccosh}\left(\sqrt{\frac{A_{stop}^2 - 1}{A_{pass}^2 - 1}}\right)}{\operatorname{arccosh}\left(\frac{\omega_{stop}}{\omega_{pass}}\right)}$$

$$= \frac{\operatorname{arccosh}\left(\sqrt{\frac{(10^{50/20})^2 - 1}{(10^{1/20})^2 - 1}}\right)}{\operatorname{arccosh}\left(\frac{8000}{2500}\right)}$$

$$= 3{,}89$$

$$\Rightarrow \underline{\underline{N = 4}}$$

b) Some challenges are:

- Complexity: High order filters lead to large and complex circuits that need careful design.
- Accuracy: Electronic components come in predefined values with tolerances. Hence, the actual filter may deviate from the original one.
- Flexibility: Analog filters can not easily be replaced/changed.

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4 a) When sampling a signal, frequencies are normalized according to

$$\Omega = \omega T_s \Leftrightarrow \omega = \frac{\Omega}{T_s}$$

which yields

$$\omega = \omega_s \Leftrightarrow \Omega = 2\pi$$

$$\omega = \frac{\omega_s}{2} \Leftrightarrow \Omega = \pi$$

$$\vdots$$

Furthermore, since $x[k]$ is a discrete-time signal, $X(\Omega)$ is 2π-periodic. This periodicity comes from sampling the continuous-time signal $x(t)$ and is thus not present in $X(\omega)$. Hence, $X(\omega)$ only consists of the component around $\Omega = 0$ (including the negative part since $X(\Omega) = X^*(-\Omega)$). From Fig. 2, we see that this part can be described by

$$X(\Omega) \propto \mathrm{rect}\left(\frac{\Omega}{2 \cdot \Omega_0}\right) \quad \text{with} \quad \Omega_0 = \frac{\pi}{4} \qquad \Omega \in (-\pi, \pi]$$

("proportional to", due to missing amplitude)

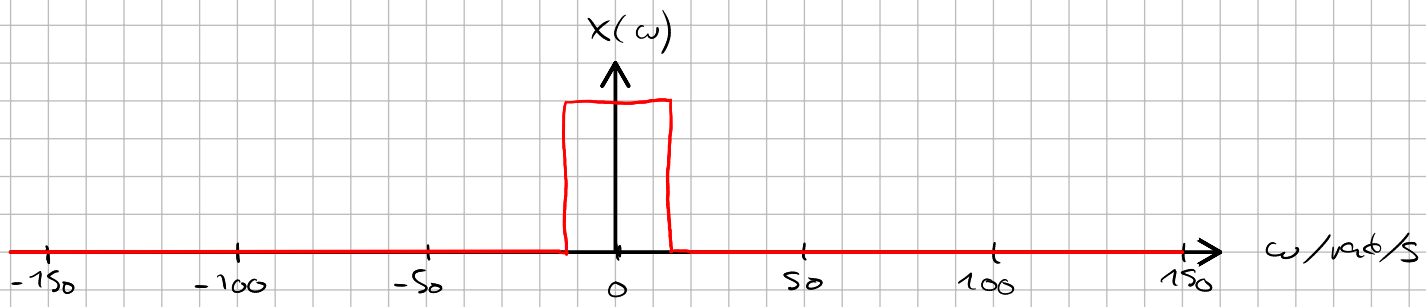
Hence, we have that

$$X(\omega) \propto \mathrm{rect}\left(\frac{\omega}{2 \cdot \omega_0}\right)$$

with

$$\omega_0 = \frac{\Omega_0}{T_s} \underset{T_s = \frac{2\pi}{\omega_s}}{=} \frac{\pi/4}{2\pi/100} = \frac{100}{8} = 12{,}5 \text{ rad/s}$$

This yields:



b) $y[k] = x[k] \cos\left(\frac{\pi}{2} k\right)$

$$\Updownarrow$$

$$Y(\Omega) = \frac{1}{2\pi} X(\Omega) * \mathcal{F}\left\{\cos\left(\frac{\pi}{2} k\right)\right\} \qquad \Omega \in (-\pi, \pi]$$

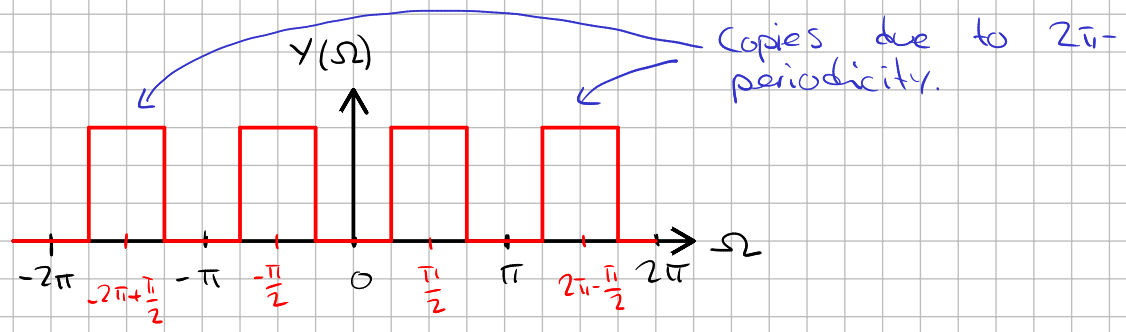
$$= \frac{1}{2\cancel{\pi}} X(\Omega) * \cancel{\pi}\left(\delta(\Omega + \pi/2) + \delta(\Omega - \pi/2)\right)$$

Convolution is on the interval

$-\pi \ldots \pi$

$$= \frac{1}{2}\left(X(\Omega+\pi/2) + X(\Omega-\pi/2)\right) \qquad \Omega \in (-\pi, \pi]$$

$$\propto \frac{1}{2}\left(\mathrm{rect}\left(\frac{\Omega+\pi/2}{2\pi/4}\right) + \mathrm{rect}\left(\frac{\Omega-\pi/2}{2\pi/4}\right)\right) \qquad \Omega \in (-\pi, \pi]$$



c) From a), we have that

$$X(\Omega) \propto \mathrm{rect}\left(\frac{\Omega}{2\pi/4}\right)$$

$$\Updownarrow$$

$$x[k] \propto \frac{\pi/4}{\pi}\,\mathrm{sinc}\left(\frac{\pi/4\ k}{\pi}\right) = \frac{1}{4}\,\mathrm{sinc}\left(\frac{k}{4}\right)$$

This shows that $x[k]$ is neither periodic, nor time-limited, which means that we would need $K \to \infty$ of samples for the best possible result with the DFT.

d) Windowing can reduce leakage compared to time-limitation ($\hat{=}$ rectangular window). However, this is achieved at the expense of a worse spectral resolution (if the number of samples is kept constant). This means that it is easier to analyze components of the signal that are sufficiently apart as leakage from nearby components causes less disturbance.

5. a) We know that

$$H(z) = \frac{1+\sqrt{2}z^{-1}+z^{-2}}{1-0{,}4z^{-1}+0{,}2z^{-2}} = \frac{Y(z)}{X(z)}$$

$$\Rightarrow \quad Y(z)(1-0{,}4z^{-1}+0{,}2z^{-2}) = X(z)(1+\sqrt{2}z^{-1}+z^{-2})$$

$$Y(z)-0{,}4z^{-1}Y(z)+0{,}2z^{-2}Y(z) = X(z)+\sqrt{2}z^{-1}X(z)+z^{-2}X(z)$$

$$\Updownarrow$$

$$\underline{\underline{y[k]-0{,}4y[k-1]+0{,}2y[k-2] = x[k]+\sqrt{2}x[k-1]+x[k-2]}}$$

b) We first need to rewrite $H(z)$ to positive form:

$$H(z) = \frac{1+\sqrt{2}z^{-1}+z^{-2}}{1-0{,}4z^{-1}+0{,}2z^{-2}} = \frac{z^{-2}(z^2+\sqrt{2}z+1)}{z^{-2}(z^2-0{,}4z+0{,}2)} = \frac{z^2+\sqrt{2}z+1}{z^2-0{,}4z+0{,}2}$$

Zeros:

$$z^2+\sqrt{2}z+1 = 0$$

pq-formula

$$z_{1,2} = -\frac{\sqrt{2}}{2} \pm \sqrt{\frac{\sqrt{2}^2}{4}-1}$$

$$= -\frac{1}{\sqrt{2}} \pm \sqrt{\frac{2}{4}-1}$$

$$= -\frac{1}{\sqrt{2}} \pm \sqrt{-\frac{1}{2}}$$

$$\underline{z_{1,2} = -\frac{1}{\sqrt{2}} \pm j\frac{1}{\sqrt{2}}}$$

Poles:

$$z^2-0{,}4z+0{,}2 = 0$$

$$p_{1,2} = +\frac{0{,}4}{2} \pm \sqrt{\frac{(-0{,}4)^2}{4}-0{,}2}$$

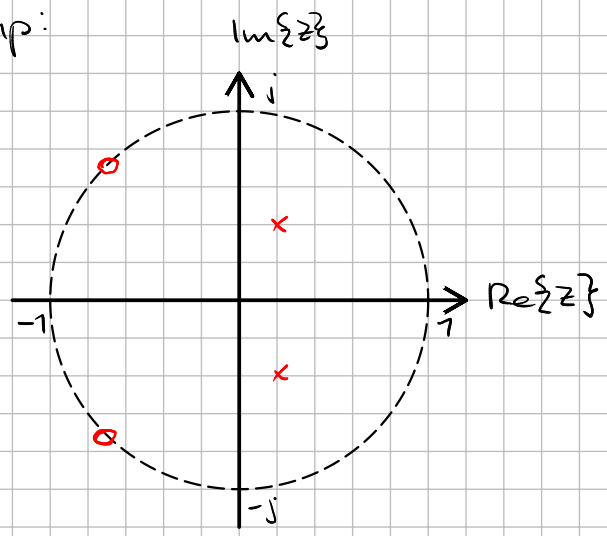
$$= 0{,}2 \pm \sqrt{\frac{0{,}16}{4}-0{,}2}$$

$$= 0{,}2 \pm \sqrt{0{,}04-0{,}2}$$

$$= 0{,}2 \pm \sqrt{-0{,}16}$$

$$\underline{p_{1,2} = 0{,}2 \pm j0{,}4}$$

Pole-zero map:



$\Rightarrow$ <u>The system is stable since $|p_i| < 1 \ \forall \ p_i$.</u>

c) To determine the gain and phase, we can look at the frequency response:

$$H(e^{j\Omega}) = H(z)|_{z=e^{j\Omega}} = \frac{1+\sqrt{2}\,e^{-j\Omega}+e^{-2j\Omega}}{1-0{,}4e^{-j\Omega}+0{,}2e^{-2j\Omega}}$$

$$H(e^{j\pi/4}) = \frac{1+\sqrt{2}\,e^{-j\pi/4}+e^{-2j\pi/4}}{1-0{,}4e^{-j\pi/4}+0{,}2e^{-2j\pi/4}}$$

$$e^{-j\pi/4} = \frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}$$

$$= \frac{1+\sqrt{2}\left(\frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}}\right)-j}{1-0{,}4\left(\frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}}\right)-0{,}2j}$$

$$e^{-2j\pi/4} = e^{-j\pi/2} = -j$$

$$= \frac{1+1-j-j}{1-0{,}4\frac{1}{\sqrt{2}}+0{,}4j\frac{1}{\sqrt{2}}-0{,}2j}$$

$$= \frac{2-2j}{0{,}72+0{,}08j}$$

$$= 2{,}44+j3{,}05$$

$$\Rightarrow \underline{\underline{|H(e^{j\pi/4})| = 3{,}9, \quad \angle H(e^{j\pi/4}) = -51{,}3^\circ}}$$

d) Since $x[k]$ is causal, we can use the z-domain input-output relationship

$$Y(z) = H(z)X(z).$$

with

$$x[k] = \sin\left(\frac{3\pi}{4}k\right)u[k]$$

$$\Updownarrow$$

$$X(z) = \frac{z^{-1}\sin\left(\frac{3\pi}{4}\right)}{1-2z^{-1}\cos\left(\frac{3\pi}{4}\right)+z^{-2}}$$

$$\sin(3\pi/4) = 1/\sqrt{2}$$

$$\cos(3\pi/4) = -1/\sqrt{2}$$

$$= \frac{z^{-1}\frac{1}{\sqrt{2}}}{1+2/\sqrt{2}\,z^{-1}+z^{-2}}$$

$$= \frac{1/\sqrt{2}\,z^{-1}}{1+\sqrt{2}\,z^{-1}+z^{-2}}$$

$$\Rightarrow \quad Y(z) = \frac{\cancel{1+\sqrt{2}\,z^{-1}+z^{-2}}}{1-0{,}4z^{-1}+0{,}2z^{-2}} \cdot \frac{1/\sqrt{2}\,z^{-1}}{\cancel{1+\sqrt{2}\,z^{-1}+z^{-2}}}$$

$$= \frac{1/\sqrt{2}\,z^{-1}}{1-0{,}4z^{-1}+0{,}2z^{-2}} \qquad (1)$$

This is of the form

$$Y(z) = K\frac{az^{-1}\sin(\Omega_0)}{1-2az\cos(\Omega_0)+a^2z^{-2}} \qquad (2)$$

Comparing (1) and (2):

1. $a^2 = 0{,}2$

$$a = \sqrt{0{,}2}$$

$$a = 0{,}44$$

2. $-2a\cos(\Omega_0) = -0{,}4$

$$a\cos(\Omega_0) = 0{,}2$$

$$\cos(\Omega_0) = \frac{0{,}2}{a}$$

$$\cos(\Omega_0) = \frac{0{,}2}{0{,}44}$$

$$\Rightarrow \quad \Omega_0 = 1{,}1$$

3. $$K a \sin(\Omega_0) = 1/\sqrt{2}$$

$$K = \frac{1}{\sqrt{2}\, a \sin(\Omega_0)}$$

$$K = 1{,}75.$$

Hence:

$$Y(z) = 1{,}75 \frac{0{,}44 \sin(1{,}1) z^{-1}}{1 - 2 \cdot 0{,}44 \cdot \cos(1{,}1) z^{-1} + 0{,}2 z^{-2}}$$

$$\Updownarrow$$

$$\underline{\underline{y[k] = 1{,}75 \cdot 0{,}44^k \sin(1{,}1\, k)\, u[k]}}$$

6.a) • $x(t)$ is band-limited to 200 rad/s. To not distort $x(t)$, this frequency should thus be chosen as the passband edge frequency, $\omega_{pass} = 200$ rad/s

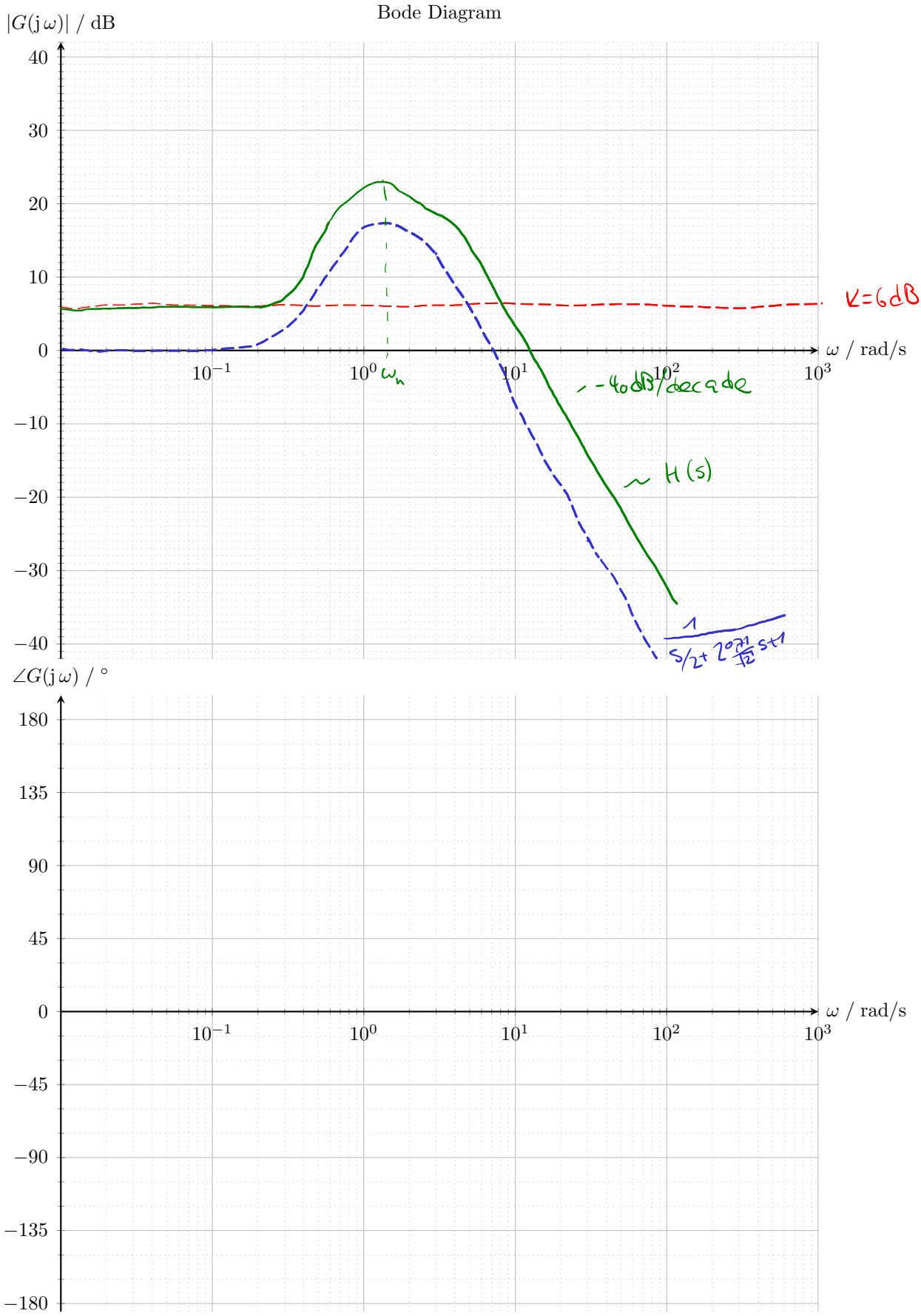
• $y(t)$ is non-zero between 600 rad/s and 1000 rad/s. Hence, to make sure $y(t)$ is suppressed it should entirely be in the stopband. This gives $\omega_{stop} = 600$ rad/s.

• The stopband edge frequency should correspond to the Nyquist frequency to avoid any form of aliasing, $\omega_N = 600$ rad/s $\Rightarrow \omega_s = 2\omega_N = 1200$ rad/s.

$$\Rightarrow \underline{\underline{\omega_{pass} = 200 \text{ rad/s}, \ \omega_{stop} = 600 \text{ rad/s}, \ \omega_s = 1200 \text{ rad/s}}}$$

b) Aliasing occurs when the sampling frequency is chosen less than twice the highest frequency component in the signal. It causes frequency components above the Nyquist frequency ($\omega_s/2$) to "fold" to lower frequencies, hence corrupting these components. A high frequency component might thus suddenly appear as a low

frequency component.


Bode Diagram
|G(j ω)| / dB
40
30
20
10
0
−10
−20
−30
−40
ω / rad/s
10^−1
10^0
10^1
10^2
10^3
ω_n
K=6dB
-40dB/decade
H(s)
1
s/2 + 2·0.71/√2 s + 1
∠G(j ω) / °
180
135
90
45
0
−45
−90
−135
−180
ω / rad/s
10^−1
10^0
10^1
10^2
10^3